Performance Prediction for Passive Localization using Conditional Normalizing Flows

Hanyue Guo*†, Rui Zhou†, Wenqiang Pu†, Licheng Zhao†, Ming-Yi You‡, JunKun Yan*§
*Hangzhou Institute of Technology, Xidian University, Hangzhou, China
†Shenzhen Research Institute of Big Data, The Chinese University of Hong Kong, Shenzhen, Shenzhen, China
‡National Key Laboratory of Electromagnetic Space Security, Jiaxing, China
§National Key Laboratory of Radar Signal Processing, Xi'an, China
Emails: 23021211826@xidian.stu.edu.cn, {rui.zhou, wpu, zhaolicheng}@sribd.cn, youmingyi@126.com, jkyan@xidian.edu.cn

***Abstract*—As 6G networks evolve, radio-based passive sensing and localization have become integral components of intelligent wireless communication systems. Numerous studies have aimed to predict localization error using specific performance metrics, such as root mean square error and the 95th percentile error, to accommodate various application scenarios. However, localization performance metrics often vary depending on the application context and user preferences. Each metric typically requires dedicated prediction models or extensive mathematical derivations, making adaptation to different scenarios and preferences challenging. To address these limitations, this paper proposes a novel error distribution modeling framework for passive localization systems based on a modified Conditional Normalizing Flow (CNF). Specifically, the framework incorporates a hierarchical structure to capture geometric characteristics and integrates environmental parameters as conditional inputs to the CNF network. Numerical experiments demonstrate that our proposed model can accurately approximate the distribution of localization errors, thereby enabling convenient and flexible prediction of localization errors under various conditional parameters.**

***Index Terms*—passive localization, conditional normalizing flows, performance prediction.**

I. Introduction

Passive localization has emerged as a key technology in radar systems and is increasingly regarded as an important component of integrated sensing and communication frameworks, attracting significant attention in recent years. [1]–[3]. By processing signals emitted by targets, it enables efficient localization with notable advantages, including high concealment and low power consumption [3]. To ensure the reliability of passive localization systems, accurate and comprehensive performance evaluation is essential.

In practical applications, the evaluation of localization performance typically relies on a variety of metrics tailored to specific scenario requirements. For example, the mean error (ME) reflects the overall level of error [4], while the root mean square error (RMSE) places greater emphasis on sensitivity to larger errors and overall system performance [5]. In safety-critical or extreme scenarios, such as emergency rescue or autonomous driving, metrics such as the maximum error or high confidence indicators (e.g., the 95th percentile error) are particularly important for ensuring reliability under worst case conditions [6] [7]. However, methods like the Cramer-Rao Lower Bound (CRLB) [8] or neural network-based RMSE modeling [9] are typically limited to providing specific quantitative metrics. These approaches have significant limitations: the introduction of each new performance metric often necessitates the redesign of dedicated modeling or estimation procedures, resulting in substantial redundant effort. Therefore, by modeling the probability distribution of localization errors, various performance metrics can be flexibly derived from a unified probabilistic framework, enabling comprehensive and consistent evaluation across diverse scenarios and requirements.

Although modeling the error distribution is crucial for performance evaluation, existing methods remain inadequate in this regard. Traditional approaches often rely on Monte Carlo (MC) methods [10]–[12], which involve repeated sampling under given conditions and approximating the distribution using techniques such as histograms [13] or kernel density estimation (KDE) [14]. While histograms can accurately capture the distribution shape when the sample size is sufficiently large, they require a substantial number of samples and are thus computationally expensive. In contrast, KDE can provide smoother estimates with relatively fewer samples, but the results are still dependent on sample quality; when the sample size is insufficient, the estimation variance increases significantly. It should be noted that each sample typically corresponds to a complete execution of the localization algorithm, making the sampling process itself computationally intensive in complex scenarios. As a result, current methods generally suffer from insufficient flexibility and low efficiency in error distribution modeling.

To address these challenges, this paper introduces a Conditional Normalizing Flow (CNF) framework for modeling the error distribution in passive localization systems. This approach enables the flexible derivation of various evaluation metrics from the learned distribution while reducing computational overhead. The main contributions are as follows:

This work was supported by the Longgang District Special Funds for Science and Technology Innovation under Grant LGKCSDPT2025002. *(Corresponding author: Rui Zhou.)*

• We propose a model based on CNF, termed the Structured Localization Error Network (SLENet), to model the probability distribution of localization errors. The proposed framework enables explicit modeling of complex localization error distributions while preserving differentiability.

• The Geometric Dilution of Precision (GDOP) is introduced as a physically interpretable representation of geometric configuration parameters and is explicitly incorporated into the error modeling process. This design embeds system geometry directly into the probabilistic model, enhancing both interpretability and modeling fidelity.

• Extensive numerical experiments demonstrate that, compared to other methods, SLENet achieves real-time and accurate error distribution prediction, and delivers superior performance across multiple evaluation metrics.

II. System Model

In this section, we briefly introduce the signal model, and then discuss the classic methods for modeling the distribution and their limitations.

A. Signal Model

Consider a passive localization scenario with N receivers are located at positions $\mathbf{p}_i^r = [x_i^r, y_i^r]$ for $i = 1, \ldots, N$. The source is located at $\mathbf{p}^s = [x^s, y^s]$. The received signal at the i-th receiver can be modeled as

$$r_i[t] = h_i s[t - \tau_i] + n_i[t], \quad i = 1, \ldots, N, \tag{1}$$

where h_i is the complex channel gain between the source and receiver i, $s[t]$ is the transmitted signal at discrete time t, τ_i is the propagation delay from the source to receiver i, and $n_i[t]$ is the additive noise at receiver i, modeled as zero-mean Gaussian noise. Time Difference of Arrival (TDOA) is a widely used technique in passive localization. By taking the difference between the arrival times of the signal at pairs of receivers, the TDOA between receiver i and a reference receiver. The set of TDOA measurements is then used to estimate the source position $\hat{\mathbf{p}}^s$ by solving a set of nonlinear equations, often via maximum likelihood methods [15].

In practical scenarios, localization estimates are affected by various sources of error. The localization error is defined as: $\epsilon = \|\mathbf{p}^s - \hat{\mathbf{p}}^s\|$. Our goal is to model the conditional probability distribution of the localization error, denoted as:

$$p(\epsilon \mid \mathbf{c}), \tag{2}$$

where $\mathbf{c} = (P,T,f_b,\mathbf{p}^s,\{\mathbf{p}_i^r\}\ )$ denotes the vector of condition variables, which includes the transmit power P, observation time T, baud rate f_b, the source coordinates $\mathbf{p}^s$, and the coordinates of receivers $\mathbf{p}_i^r$. The relationship between system parameters and localization errors is often highly nonlinear and complex. Moreover, the estimation process is affected by various sources of noise, making it challenging to derive an analytical expression for the conditional error distribution $p(\epsilon \mid \mathbf{c})$. Therefore, most existing studies rely on approximate methods to obtain the localization error distribution.

B. Traditional Methods for Distribution Modeling

The traditional distribution modeling typically relies on MC methods, which estimate probability distributions through repeated sampling. Specifically, under given conditions $\mathbf{c}$, the localization algorithm is repeatedly executed to generate M independent error samples $\{\epsilon_j\}_{j=1}^{M}$, from which the error distribution is subsequently modeled using various statistical techniques. Traditional approaches include:

1) Histogram Estimation: Histogram estimation is one of the most intuitive and straightforward methods for probability density modeling [13]. In this approach, the range of error samples is divided into several bins of fixed width $\Delta\epsilon$. The number of samples falling into each bin is counted and then normalized to approximate the probability density:

$$\hat{p}_{\text{hist}}(\epsilon_b \mid \mathbf{c}) = \frac{n_b}{M \cdot \Delta\epsilon}, \tag{3}$$

where n_b is the number of samples in the b-th bin and ϵ_b is the bin center. Although this method is easy to implement, the estimated probability density function is highly sensitive to the choice of interval width, and performs poorly with limited sample sizes. Generating each sample typically requires a complete and computationally intensive run of the localization algorithm. When system conditions change, new samples must be collected and the distribution re-estimated, which limits the adaptability and real-time applicability of this approach.

2) Kernel Density Estimation: KDE constructs a smooth, continuous, and differentiable probability density estimate by superimposing kernel functions at each sample location [16]. KDE is defined as

$$\hat{p}_{\text{KDE}}(\epsilon \mid \mathbf{c}) = \frac{1}{Mh} \sum_{j=1}^{M} K\left(\frac{\epsilon - \epsilon_j}{h}\right), \tag{4}$$

where $K(\cdot)$ is usually a Gaussian kernel and h is the bandwidth parameter controlling the degree of smoothing. KDE reduces the strict dependence on large sample sizes by introducing a smoothing kernel function. However, when the number of available samples is insufficient, the quality of the estimation is significantly affected by the sample quality, resulting in high variance in the KDE estimates. This limitation directly undermines the accuracy and stability of the final density estimation, thereby impacting the reliability of KDE.

To address these challenges, we introduce SLENet, a method based on CNF that directly learns the overall structure of the error distribution from data. Once trained, SLENet enables direct distribution prediction in the inference stage, eliminating the need for computationally expensive repeated MC sampling and thereby achieving substantially higher inference efficiency.

III. Error distribution modeling using Conditional Normalizing Flow

By incorporating system parameters as conditional inputs, CNF enables flexible represent complex passive localization error distributions. The core idea is to transform a simple base distribution into a complex target distribution through a sequence of invertible and differentiable mappings f, where the transformations are conditioned on $\mathbf{c}$ [17]. The target distribution is obtained from the basic variable change formula:

$$p(\epsilon \mid \mathbf{c}) = p_u(z) \left| \det \frac{\partial z}{\partial \epsilon} \right|, z = f^{-1}(\epsilon, \mathbf{c}), \tag{5}$$

where $p_u(z)$ denotes the base distribution, and $z = f^{-1}(\epsilon, \mathbf{c})$ represents the latent variable obtained by applying the conditional inverse transformation to the error variable ϵ. The term $\frac{\partial z}{\partial \epsilon}$ is the Jacobian matrix of the inverse transformation with respect to ϵ, whose determinant reflects the change in probability density volume induced by the transformation. In practice, these transformations are carefully designed to be both easily invertible and efficient in computing the Jacobian determinant, thereby ensuring the efficiency of the model during both training and inference [18].

A. Preliminary Approach: The vanilla CNFNet Model

A standard normal distribution is used as a fixed latent prior in our model. The normalizing flow network is constructed by stacking multiple invertible transformation modules. Each module consists of an autoregressive rational quadratic spline (RQS) flow layer [19], which provides highly flexible modeling of complex distributions, followed by an LU linear permutation layer to enhance feature mixing and the overall expressive power of the model. All flow layers are conditioned on a system feature vector, enabling the model to adaptively adjust the error distribution in response to varying environments and configurations. Specifically, the feature vector is defined as $\mathbf{c} = (P,T,f_b,\mathbf{p}^s,\{\mathbf{p}_i^r\}$), where these variables represent the relevant system parameters. For brevity, this network is referred to as CNFNet. To account for the varying scales of localization error, we propose predicting the logarithm of the localization error. During training, the model is optimized by minimizing the forward Kullback-Leibler (KL) divergence between the true error distribution and the model's estimated distribution [20]. Once training is complete, the CNF model can efficiently compute the log-probability density and various statistical metrics of the errors under new system conditions, without the need for expensive Monte Carlo simulations.

B. The Proposed SLENet Model

In the input feature set, the positions of the five receiving stations contribute 12 dimensions for a single target scenario. The localization performance is not determined by the coordinates of individual receiving stations in isolation, but rather by the overall geometric configuration formed jointly by all receivers. To enhance the geometric interpretability of the model and reduce its complexity, we map these positional features to a single geometric accuracy metric: the GDOP, which quantifies the effect of spatial geometry on localization accuracy. Through this transformation, the dimensionality of the input features is significantly reduced, resulting in fewer model parameters, a lower risk of overfitting. For the TDOA localization method, GDOP is defined as

TABLE I: Data generation parameters in $\mathcal{D}$.

Feature	Value
Sample Rate	10 MHz
Transmit Power	$[3, 15]$ W
Observation Time	$[0.5, 10]$ ms
Baud Rate	$f_s/[16,20,25,32,40,50,64,80,100]$ kHz
Target	$x^s, y^s \in [0, 30]$ km
Receiving stations	$x_i^r, y_i^r \in [0, 30]$ km

$$\text{GDOP}(\mathbf{p}^s, \{\mathbf{p}_i^r\}) = \sqrt{\text{tr}((\mathbf{C}^T\mathbf{C})^{-1})} \tag{6}$$

where $\mathbf{C}$ is the geometry matrix that encapsulates the effects of the spatial layout on positioning accuracy [21] [22]. Apart from this modification to the input representation, the remaining network architecture and training procedure are identical to those of CNFNet. In summary, modeling the distribution of localization errors by incorporating the vector $\mathbf{c} = (P,T,f_b,\text{GDOP}$) as the network input can significantly enhance the model's adaptability to changes in system configuration and the accuracy. We refer to this model as SLENet.

IV. Numerical Experiments

In this section, we describe the data generation process and evaluate the performance of SLENet on the TDOA method through numerical experiments.

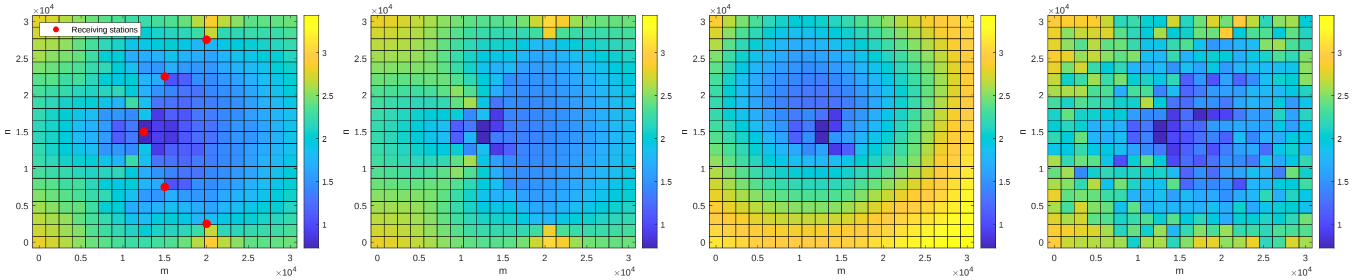
A. Dataset Generation

For simplicity, we consider a two-dimensional scenario with five receiving stations; the proposed method can be readily extended to three-dimensional cases. The source signals signals are modulated using binary phase-shift keying and generated via the MATLAB Communications Toolbox. A dataset $\mathcal{D}$ is constructed using the TDOA method, with each data point labeled through 500 Monte Carlo simulations. The feature vector $\boldsymbol{\theta}$ consists of 15 features: the source's transmitting power (P), observation time (T), baud rate (f_b), and the two-dimensional coordinates of both the radiated source and the five receiving stations. The final dataset $\mathcal{D}$ contains 52000 data points, with parameters uniformly and randomly sampled from predefined ranges, as detailed in Table I.

B. Performance Prediction

The SLENet model consists of $K = 20$ stacked transformation blocks. Each RQS transformation contains two hidden layers with 64 units per layer. The model is trained using the Adam optimizer with an initial learning rate of 0.005, a batch size of 1024, and for a total of 200 epochs. During training, the Kullback Leibler divergence loss is monitored on both the training and validation sets. The model parameters corresponding to the lowest validation loss are saved for testing and further analysis. The architecture of CNFNet is identical to that described above.

To ensure robust model evaluation, dataset $\mathcal{D}$ is partitioned into training, validation, and test sets in a 6 : 3 : 1 ratio. Table II summarizes the evaluation results of various methods on the test set. KL uses a 100-bin histogram as the empirical



(a) Ground Truth. (b) SLENet. (c) CNFNet. (d) Gaussian KDE.

Fig. 1: The true and predicted (log-scaled) 95th percentile error map with parameters: $P = 7$W, $T = 0.2$ms, $f_b = 400$kHz.

TABLE II: Performance evaluation metrics of different methods on the test dataset.

	KL	WD	ME MAE	RMSE MAE	95% MAE
SLENet (Prop.)	**1.6073**	**0.1240**	**0.1099**	**0.0731**	**0.1164**
CNFNet	1.8080	0.1958	0.2313	0.1617	0.2990
Gaussian KDE (MC=10)	2.3590	0.1469	0.1186	0.0772	0.1781

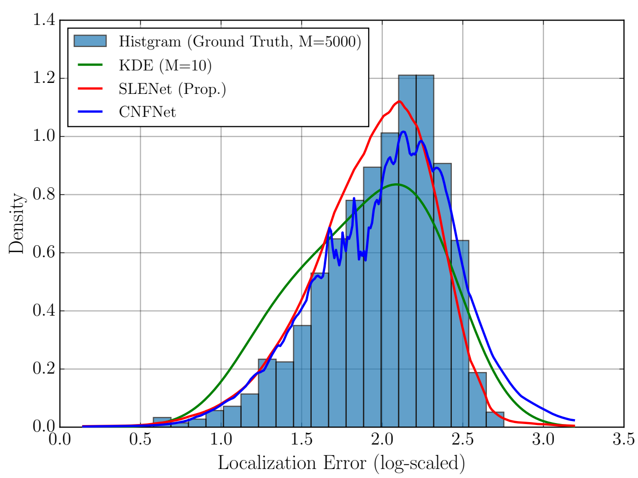


Fig. 2: Comparison of localization error distributions.

distribution, while Wasserstein samples from the estimated grid distribution with the same size as the observed errors. Among the reported metrics, KL divergence and Wasserstein distance (WD) quantify the differences between the predicted and true distributions. The remaining performance metrics are derived from the predicted error distributions, including the ME, RMSE, and the 95th percentile error. Estimation accuracy is evaluated by measuring the mean absolute error (MAE) between the predicted values and the ground truth. The results indicate that SLENet consistently outperforms other methods across all metrics, demonstrating superior capabilities in distribution fitting and error estimation. Conversely, the KDE method, which relies on repeated data generation for different conditions, produces unreliable estimates with limited sample sizes and is therefore unsuitable for real-time applications.

Fig. 1 illustrates the 95th percentile error maps generated by different methods when only the target location varies, with all other conditions held constant. The ground truth 95th percentile error map, shown in 1a, is obtained from $5,000$ Monte Carlo simulations and serves as the reference for evaluation. Fig. 1b shows the error map predicted by SLENet, which closely matches the ground truth and achieves a MAE of 0.1198. In comparison, the CNFNet and KDE methods yield higher MAE values of 0.5370 and 0.1839, respectively. These results indicate that SLENet achieves the best performance in both visual appearance and MAE evaluation.

Furthermore, Fig. 2 presents an intuitive comparison of different methods by illustrating their ability to fit the localization error distribution under identical parameter conditions. The histogram uses 20 bins, density is evaluated on a 100-point uniform grid, and KDE uses a Gaussian kernel with bandwidth chosen by Scott's rule. The results clearly demonstrate that SLENet provides the closest approximation to the true error distribution. This improvement is primarily attributed to effective feature dimensionality reduction, which allows the network to more efficiently learn the relationship between spatial geometry and localization errors, thereby enhancing performance across a wide range of scenarios and parameter settings. In terms of computational efficiency, experiments conducted on a Lenovo ThinkSystem SR630 server without parallel acceleration show that the distribution prediction times for KDE with MC $= 10$, MC $= 5000$, and SLENet are 26.97, 12639.27, and 0.1 seconds, respectively, demonstrating the superior computational efficiency of the proposed method.

V. Conclusion

This paper proposes SLENet, a model based on Conditional Normalizing Flows for modeling localization error distributions in passive localization systems. Using GDOP as a key feature, SLENet achieves significant improvements in accuracy and flexibility over traditional methods. Furthermore, its unified probabilistic framework enables flexible derivation of multiple performance metrics, supporting comprehensive evaluation across diverse scenarios.

REFERENCES

[1] N. Jabeen, H. Lei, A. Muhammad, A. Ali, Z. U. Khan, and G. Pan, "Localization in ISAC: A review," *IEEE Internet of Things Journal*, 2025.

[2] P. Wan, Y. Ni, B. Hao, Z. Li, and Y. Zhao, "Passive localization of signal source based on wireless sensor network in the air," *International Journal of Distributed Sensor Networks*, vol. 14, no. 3, p. 1550147718767371, 2018.

[3] Q. Zhou, Y. Yuan, H. Li, Y. Lai, M. S. Greco, F. Gini, and W. Yi, "Target localization for distributed hybrid active-passive radars," *IEEE Transactions on Aerospace and Electronic Systems*, 2025.

[4] A. S. Lutakamale, H. C. Myburgh, and A. De Freitas, "Rssi-based fingerprint localization in lorawan networks using cnns with squeeze and excitation blocks," *Ad hoc networks*, vol. 159, p. 103486, 2024.

[5] M. Z. Win, A. Conti, S. Mazuelas, Y. Shen, W. M. Gifford, D. Dardari, and M. Chiani, "Network localization and navigation via cooperation," *IEEE Communications Magazine*, vol. 49, no. 5, pp. 56–62, 2011.

[6] S. S. Afzal, W. Chen, and F. Adib, "3d-blue: Backscatter localization for underwater robotics," in *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2024, pp. 11 040–11 047.

[7] G. K. Fischer, T. Schaechtle, M. Schabinger, A. Richter, I. Häring, F. Höflinger, and S. J. Rupitsch, "Massloc: A massive sound source localization system based on direction-of-arrival estimation," *arXiv preprint arXiv:2508.12024*, 2025.

[8] M. Wylie and J. Holtzman, "The non-line of sight problem in mobile location estimation," in *Proceedings of ICUPC - 5th International Conference on Universal Personal Communications*, vol. 2, 1996, pp. 827–831 vol.2.

[9] H. Guo, R. Zhou, W. Pu, M.-Y. You, and J. Yan, "Using monotonic neural networks for accurate and efficient passive localization performance modeling," in *2025 IEEE Wireless Communications and Networking Conference (WCNC)*. IEEE, 2025, pp. 1–6.

[10] D. Fox, W. Burgard, F. Dellaert, and S. Thrun, "Monte Carlo localization: Efficient position estimation for mobile robots," *Aaai/iaai*, vol. 1999, no. 343-349, pp. 2–2, 1999.

[11] F. Gustafsson and F. Gunnarsson, "Positioning using time-difference of arrival measurements," in *2003 IEEE International Conference on Acoustics, Speech, and Signal Processing, 2003. Proceedings. (ICASSP '03).*, vol. 6, 2003, pp. VI–553.

[12] C.-C. Hsu, C.-J. Kuo, and W.-C. Kao, "Improved Monte Carlo localization with robust orientation estimation for mobile robots," in *2013 IEEE International Conference on Systems, Man, and Cybernetics*, 2013, pp. 3651–3656.

[13] S. M. Pizer, E. P. Amburn, J. D. Austin, R. Cromartie, A. Geselowitz, T. Greer, B. ter Haar Romeny, J. B. Zimmerman, and K. Zuiderveld, "Adaptive histogram equalization and its variations," *Computer vision, graphics, and image processing*, vol. 39, no. 3, pp. 355–368, 1987.

[14] J. Heer, "Fast & accurate gaussian kernel density estimation," in *2021 IEEE Visualization Conference (VIS)*. IEEE, 2021, pp. 11–15.

[15] Y. Chan and K. Ho, "A simple and efficient estimator for hyperbolic location," *IEEE Transactions on Signal Processing*, vol. 42, no. 8, pp. 1905–1915, 1994.

[16] G. R. Terrell and D. W. Scott, "Variable kernel density estimation," *The Annals of Statistics*, pp. 1236–1265, 1992.

[17] C. Winkler, D. Worrall, E. Hoogeboom, and M. Welling, "Learning likelihoods with conditional normalizing flows," *arXiv preprint arXiv:1912.00042*, 2019.

[18] G. Papamakarios, E. Nalisnick, D. J. Rezende, S. Mohamed, and B. Lakshminarayanan, "Normalizing flows for probabilistic modeling and inference," *Journal of Machine Learning Research*, vol. 22, no. 57, pp. 1–64, 2021.

[19] C. Durkan, A. Bekasov, I. Murray, and G. Papamakarios, "Neural spline flows," *Advances in neural information processing systems*, vol. 32, 2019.

[20] L. Dinh, D. Krueger, and Y. Bengio, "Nice: Non-linear independent components estimation," *arXiv preprint arXiv:1410.8516*, 2014.

[21] D. J. Torrieri, "Statistical theory of passive location systems," *IEEE transactions on Aerospace and Electronic Systems*, no. 2, pp. 183–198, 1984.

[22] M. Wang, Z. Chen, Z. Zhou, J. Fu, and H. Qiu, "Analysis of the applicability of dilution of precision in the base station configuration optimization of ultrawideband indoor tdoa positioning system," *IEEE Access*, vol. 8, pp. 225 076–225 087, 2020.